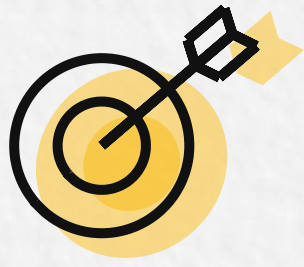


Foundational Content- Learning Map



Goal:

Build the core Python and Agentic AI foundational knowledge required to confidently understand LLMs, and AI agents before the program begins.

Overview...

Applied Agentic AI for SWEs - Foundational Material - 2026

This foundational content is designed to support learners from different backgrounds. It ensures everyone has the essential Python knowledge needed to follow live sessions, complete hands-on work, build capstone projects and effectively understand and work with Large Language Models (LLMs) and AI agents.

Before Class 1, learners are required to complete the Mandatory Python Resources. Optional Deep-Dive Resources are available for those who want additional conceptual depth and extended practice and can be explored at their own pace during the course.



1

Mandatory Python Resources:

Introduction to Python
Data Types & Strings
Strings, Lists, & Tuples
Sets & Dictionaries
Conditionals (If-Else Statements)
Loops
Functions
Modules & Packages
Pandas (Core Usage)

2

Optional Resources:

Extended Python Foundations
Evolution of Generative AI
Basics of Prompt Engineering
Basics of Text Processing

Mini Project:
Hello Agent – CSV FAQ
Agent

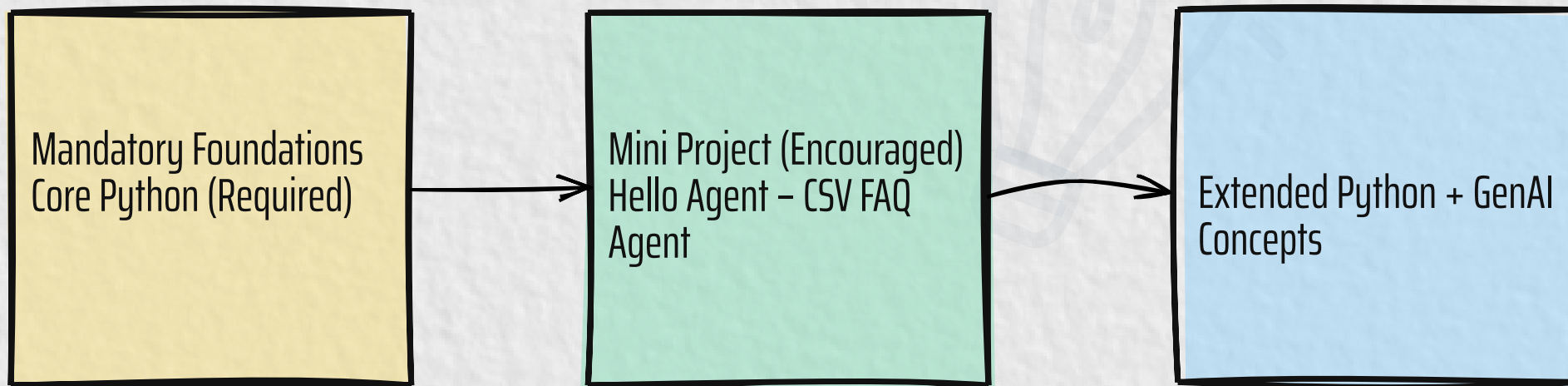
Note:

Focus on completing the Mandatory Python Resources before Week 1. Everything else is optional and meant to support you based on your background and learning goals

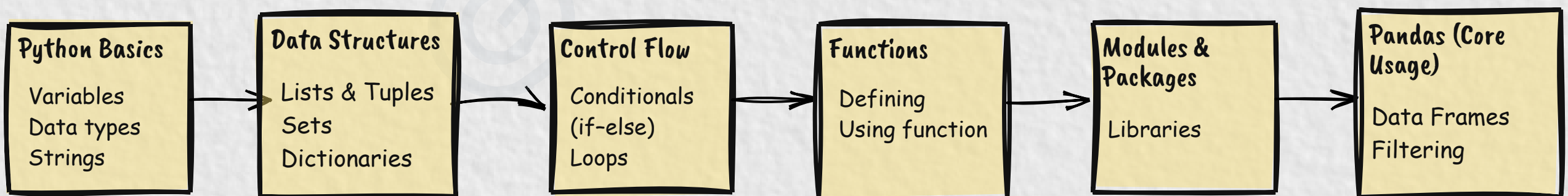
If you have questions about which resources to follow, please reach out to the Operations team for guidance.



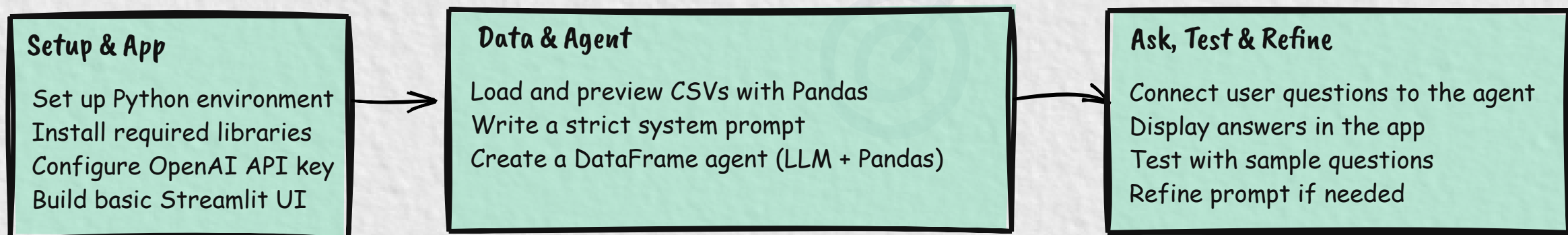
Overview



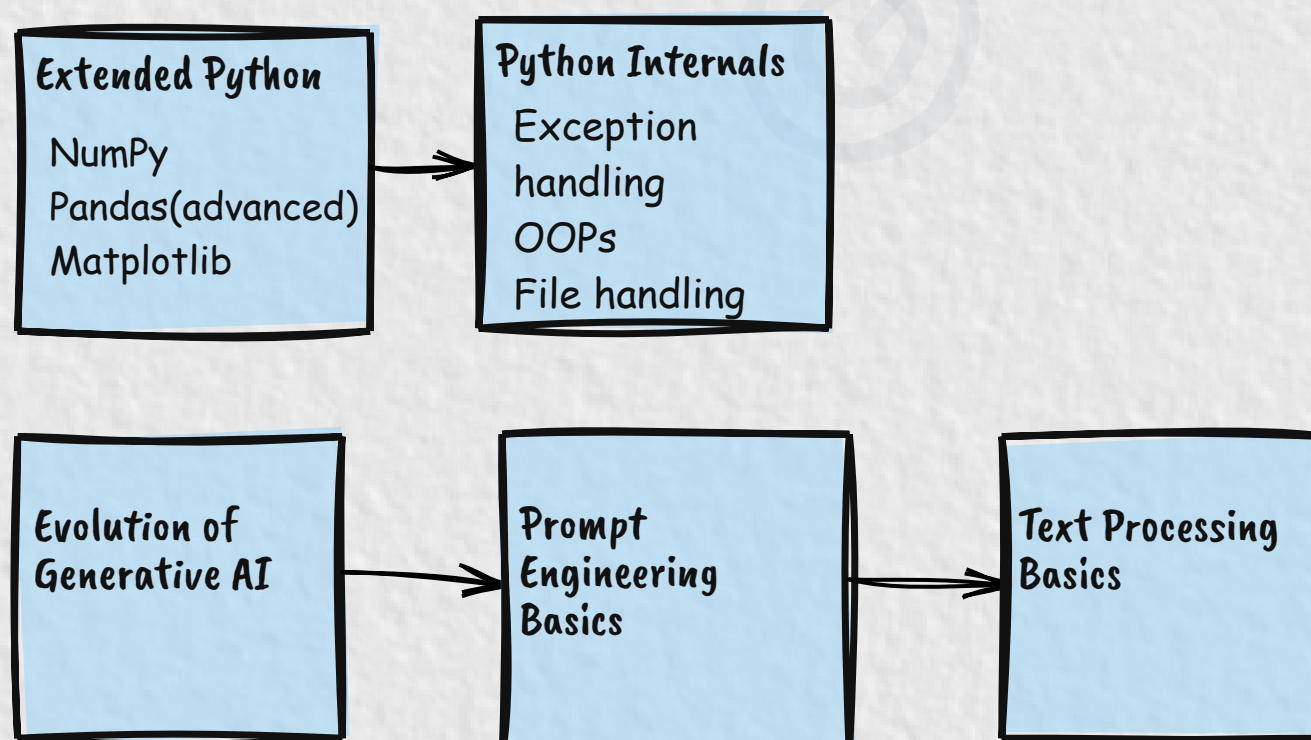
1 Mandatory Python Roadmap (Required)



2 Mini Project - Hello Agent (CSV FAQ Agent)



3 Optional Content



For more details on the topics covered in each resource, please refer to the detailed Foundational Material Learner Guide on your Uplevel account.